PRG 107 – Python Programming Kristiania University of Applied Sciences Lec 10

By Huamin Ren

Huamin.ren@kristiania.no

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Outline – Python in data science: I

1. Introducing Pandas Objects
2. Data Indexing and Selection
3. Operating on Data in Pandas
4. Handling Missing Data

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Introduction to Pandas

- Pandas is a newer package built on top of NumPy, and provides an efficient implementation of a DataFrame
- Important three fundamental Pandas data structures
  - Series
  - DataFrame
  - Index

1.1 Series

- A Pandas Series is a one-dimensional array of indexed data.
- Atributes: Values and Index

```
# 1.1
data = pd.Series([0.25, 0.5, 0.75, 1.0])
data
```

```
0    0.25
1    0.50
2    0.75
3    1.00
dtype: float64
```

```
data.values
data.index
```

```
RangeIndex(start=0, stop=4, step=1)
```

- Data can be accessed by the associated index via the familiar Python square-bracket notation

```
print(data[1])
print(data[1:3])
```

```
0.5
1    0.50
2    0.75
dtype: float64
```

- The essential difference: while the Numpy Array has an *implicitly defined* integer index used to access the values, the Pandas Series has an *explicitly defined* index associated with the values.

- This explicit index definition gives the Series object additional capabilities.

For example, the index need not be an integer, but can consist of values of *any desired* type.

```
11 # 1.1.2
   data = pd.Series([0.25, 0.5, 0.75, 1.0],
                    index=['a', 'b', 'c', 'd'])
   data

11 a    0.25
   b    0.50
   c    0.75
   d    1.00
   dtype: float64

12 print(data['b'])

   0.5
```

- Think of a Pandas Series like a specialization of a Python dictionary.

A dictionary is a structure that maps arbitrary keys to a set of arbitrary values

A Series is a structure which maps typed keys to a set of typed values

The type information of a Pandas Series makes it much more efficient than Python dictionaries for certain operations

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Constructing Series objects

- pd.Series(data, index=index)

can be one of many entities

an optional argument

Data can be…

- A list or NumPy array
- A scalar
- A dictionary

1.2 Dataframe

- DataFrame as a generalized NumPy array
- DataFrame as specialized dictionary

DataFrame as a generalized NumPy array

- An analog of a two-dimensional array with both flexible row indices and flexible column names

	population	area
California	38332521	423967
Texas	26448193	695662
New York	19651127	141297
Florida	19552860	170312
Illinois	12882135	149995

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DataFrame as specialized dictionary

- Dictionary: maps a key to a value
- DataFrame: maps a column name to a Series of column data

```
states['area']

California    423967
Texas         695662
New York      141297
Florida       170312
Illinois      149995
Name: area, dtype: int64
```

better to think about DataFrames as generalized dictionaries rather than generalized arrays

Other constructing ways

- From a single Series object
- From a list of dicts
- From a two-dimensional NumPy array
- From a NumPy structured array (self-learning)

1.3 Index

- Index: can be thought of either as an *immutable array* or as an *ordered set*